Chapter 1 exercises

These exercises are original to this chapter. State the candidate set, the counting rule, and the final calculation. A numerical answer alone is incomplete.

1. A school course code has three distinct uppercase English letters followed by two distinct digits. Neither O nor I is permitted, and the first digit cannot be zero. How many codes exist?
2. Choose a chair, a vice chair, and three committee members from eight candidates. Each person has at most one role, and the three ordinary members have no ordered offices. How many assignments record these roles and identities?
3. How many arrangements of all letters of BANANARAMA begin and end with A?
4. Give a combinatorial proof of

$$\sum_{k=0}^{n} k\binom{n}{k} = n2^{n-1}, \quad n \geq 1.$$

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Chapter 1 exercises (continued)

5. How many nonnegative integer solutions of $x_1 + x_2 + x_3 + x_4 = 13$ have every entry at least two?
6. An admissible word has n opening and n closing parentheses, with no prefix having more closings than openings. Use Catalan numbers to count these words for $n = 6$, and explain why $\binom{12}{6}/2$ is invalid.
7. Count arrangements of BANANARAMA with no adjacent A's. Then also require A at both ends. Explain why the gap encoding is uniquely reversible.
8. A nine-bit word has six 1's and three 0's. Count words with exactly four runs and with exactly five runs. If the number of 0's becomes zero while six 1's remain, can the two-positive-count formula still be used directly?
9. Put seven distinct books on three labeled shelves with left-to-right order. Count placements with no empty shelf and placements allowing empty shelves. Compare with recording only each book's destination, and explain why the three answers do not conflict.
10. Find $[x^2y^3z^2](x - 2y + z)^7$ and the sum of all coefficients. How should the factor-selection argument be interpreted when coefficients may be negative?
11. A length-eleven step word contains seven $+1$'s and four -1's. Count words with every prefix nonnegative. State the reflection target for inadmissible words and its inverse explicitly.
12. Count ordered full binary trees with five internal nodes, triangulations of a convex heptagon, and full parenthesizations of six factors in fixed order. Explain their Catalan parameters. What changes for seven factors?
13. Count integer triples $1 \le k < j \le i \le n$, using a bijection to strictly increasing subsets, and evaluate at $n = 8$. How does this differ from the lecture's loop with both inequalities weak?
14. Count integer solutions of $x_1 + x_2 + x_3 + x_4 = 11$, with $x_1 \ge 2, x_2 \ge 1, x_3 \ge 0, 0 \le x_4 \le 3$. Remove lower bounds first, then use a complement for the single upper bound. Do not directly multiply purported fixed choices for the entries.

References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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